

A Novel Expandable Borderline Smote Over-Sampling Method for Class Imbalance Problem

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Abstract—The class imbalance problem can cause classifiers to be biased toward the majority class and inclined to generate incorrect predictions. While existing studies have proposed numerous oversampling methods to alleviate class imbalance by generating extra minority class samples, these methods still have some inherent weaknesses and make the generated samples less informative. This study proposes a novel over-sampling method named the Expandable Borderline Smote (EB-Smote), which can address the weaknesses of existing over-sampling methods and generate more informative synthetic samples. In EB-Smote, not only minority class but also majority class is oversampled, and the synthetic samples are generated in the area between the selected minority and majority samples, which are close to the borderlines of their respective classes. EB-Smote can generate more informative samples by expanding the borderlines of minority and majority classes toward the actual decision boundary. Based on 27 imbalanced datasets and commonly used machine learning models, the experimental results demonstrate that EB-Smote significantly outperforms the other 8 existing oversampling methods. This study can provide theoretical guidance and practical recommendations to solve the crucial class imbalance problem in classification tasks.

Index Terms—Class imbalance, oversampling, machine learning, classification, decision boundary.

I. INTRODUCTION

MACHINE learning models have exhibited significant advancements in recent decades, with their application becoming progressively more widespread [1]. To construct an accurate classifier model, one of the nontrivial challenges is learning from an imbalanced dataset, where one class contains a much smaller number of samples than other classes. Due to the imbalanced distribution, the classifiers are biased toward the

majority class and tend to misclassify minority class samples, causing significant concern if the correct recognition of the minority class is of key importance [2]. This problem has become notorious in numerous practical application fields, including credit default prediction [3], fraud detection [4], online review manipulation [5], and bankruptcy prediction [6]. Thus, there is a need to improve learning from imbalanced datasets.

To overcome the problem of class imbalance, prior research has introduced a number of strategies, broadly divided into cost-sensitive methods, algorithm-level methods, and data-level methods [7], [8], [9]. Cost-sensitive methods consider the varying misclassification costs of different classes in the dataset [10]. Generally, the minority class samples are assigned higher misclassification costs than the majority class to increase the weight of the minority class in the dataset. Algorithm-level approaches encompass the modification of current algorithms or the creation of new algorithms specifically designed to directly tackle datasets exhibiting class imbalance, which can lead to superior performance compared to traditional algorithms [11], [12], [13]. Data-level methods aim to rebalance the class distribution in the original datasets [14]. These methods typically involve over-sampling the minority class by generating synthetic minority samples, under-sampling the majority class by selectively eliminating a portion of them or combining both oversampling and under-sampling. Due to their straightforward interpretability and high efficiency, data-level techniques have gained prominence as a viable solution and are extensively implemented in practical contexts. Within this category, oversampling methods stand out for not only balance the class distribution of datasets but also preserving the integrity of the original information [14].

The idea of oversampling methods is to generate new informative minority samples to provide more information for classification [2], [15], [16]. Among the popular oversampling methods, random oversampling is the simplest strategy, which duplicates existing minority samples to balance the ratio between the minority and majority classes. However, this method created so many samples located in the same place and shrunk the decision boundary to multiple duplicated samples, resulting in the overfitting problem [14]. Then, the more intelligent Synthetic Minority Oversampling Technique (Smote) is proposed, and synthetic minority samples are created along the line joining each minority class sample with its random number of nearest neighbors [17]. Subsequently, a large strand of literature modified the original Smote and proposed various Smote variants,

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which mainly include two categories of modification strategies, i.e., sample selection and sample generation. Sample selection focuses on selecting more important samples as initial samples for the next generation process, such as Borderline-Smote [18], SVM-Smote [19], and Kmeans-Smote [7]. Sample generation focuses on improving the mechanism of generating new synthetic samples, such as RCSmote [20], Geometric Smote [21], and CHSmote [14]. Although Smote and the variants can alleviate the class imbalance problem to some extent, they still have some weaknesses that cause the synthetic samples to be less informative, i.e., generating samples far away from borderlines, generating noise, generating overly dense synthetic samples [14] or ignoring the external area of the minority class. The details of these weaknesses are explained in the next section.

In this study, to address the class imbalance problem, we propose a novel expandable borderline-based Smote (EB-Smote) algorithm, which can simultaneously overcome the four weaknesses of existing oversampling methods. These weaknesses are addressed by the two crucial components of EB-Smote, i.e., selection of border samples (for the first two weaknesses) and generation of expanding synthetic samples (for the last two weaknesses). In the first component, EB-Smote initially separates minority samples into border, noise and safety, where the border represents samples close to the decision boundary between minority and majority class, noise represents outliers invading the area of majority class, and safety denotes samples far away from the boundary and then select the border samples for the next generation process. The border majority samples are selected similarly. In the second component, EB-Smote creates synthetic minority and majority samples between the border minority and majority samples. Compared with synthetic samples by existing oversampling methods, the synthetic samples based on EB-Smote are closer to the decision boundary and can expand the borderlines of both minority and majority classes toward the decision boundary. Since the samples closer to the boundary can provide more information for classification [22], it is expected that EB-Smote can generate more informative synthetic samples and alleviate the class imbalance problem better.

This study has the following contributions. First, this study proposes a novel oversampling method EB-Smote to solve the class imbalance problem faced by machine learning methods. Existing studies have proposed various oversampling methods to alleviate the imbalance problem, e.g., the widely used Smote and its variants [14], [17], [18], [19], [20], [23]. However, these methods still have several weaknesses, causing the synthetic samples to be less informative. This study proposes that EB-Smote consists of two components, i.e., the selection of border samples and the generation of expanding synthetic samples, which can simultaneously address the weaknesses of existing oversampling methods.

Second, this study demonstrates that oversampling majority samples can also play an important role in alleviating the class imbalance problem. Existing oversampling methods mainly focus on increasing the number of minority class samples [13], [20], [21], [24], [25] but seldom consider generating the majority class samples. The reason seems to be that generating extra majority samples can aggravate the class imbalance. Nonetheless,

the majority samples around the borderline are equally important as minority samples around the borderline since the border samples are closer to the decision boundary and can provide more information. Thus, when oversampling the minority samples, we also adopt a similar scheme to generate a proper number of majority samples around the borderline.

II. LITERATURE REVIEW

In early studies, the main factor in the imbalanced dataset that causes the performance decrease is thought of as the global imbalance ratio between imbalanced classes (also called between-class imbalance). In addition, subsequent studies find that the decline in classification efficacy is influenced by additional aspects concerning data distribution. These include within-class imbalance, where the minority class is fragmented into numerous infrequent subconcepts acting as minor disjuncts [13], and the presence of noisy samples characterized by an excess of minority samples within regions predominantly occupied by the majority class [26], and class overlap, where minority and majority classes coexist within the same region of the data space [22]. When these factors occur together in the imbalanced dataset, recognition of the minority class becomes difficult.

To overcome the class imbalance problem, various methods have been proposed. These methods can be broadly divided into cost-sensitive methods, algorithm-level methods, and data-level methods. Cost-sensitive methods modify the training loss for different classes to mitigate the imbalanced learning effects [27]. The algorithm-level (or model-perspective) approaches involve modifying existing algorithms or developing new ones specifically designed to directly address class imbalance in datasets. For example, to optimize areas under precision-recall curves (AUPRC), Qi et al. [28] propose efficient adaptive and non-adaptive stochastic algorithms and Zhang et al. [29] develop a novel federated learning method. Similarly, Ying et al. [30] propose a stochastic online algorithm to maximize the area under the receiver operating characteristic curve (AUROC).

Data-level methods aim to rebalance the class distribution in the original datasets. Among these methods, oversampling is one of the most popular methods, which can balance the ratio between different classes by adding minority samples [13]. The rationale behind oversampling methods is to expose the classifier to both classes similarly. The most basic strategy for oversampling involves randomly duplicating samples from the minority class. Although it can help alleviate the degree of class imbalance in the dataset, simple duplication can cause the overfitting problem because these samples have the same features [19]. A more advanced oversampling strategy called Smote [17] is proposed. In Smote, the new synthetic minority class sample is randomly generated on the line segments between each minority class sample and its nearest neighbors. Compared with oversampling with replication, the Smote can fill the sparse distribution of the minority class more densely.

To better overcome the key difficulties in imbalanced datasets, a range of Smote derivatives have been developed through the integration of additional data preprocessing methodologies or alterations to the existing Smote algorithms [20], [27]. These

adaptations primarily fall into two strategic categories: sample selection and sample generation [14].

Sample selection refers to the procedure of choosing initial minority samples for subsequent generation. This involves assessing the significance of various minority samples, thereby facilitating the selection of pivotal ones to produce samples rich in information. Borderline-Smote, proposed by Han et al. [18], focuses on selecting samples around the borderline as the initial samples. Samples from minority classes situated near the boundary separating two classes are more susceptible to classification errors, making them more critical for accurate classification compared to those located further away from such dividing lines. Similarly, [19] proposed a borderline oversampling method SVM-Smote, in which the qualified minority samples around the borderline are the support vectors obtained by a trained SVM. Moreover, [23] presented a novel adaptive synthetic (Adasyn) to use a density distribution for different minority class samples according to their level of difficulty in classification. Minority class examples that are harder to learn will be assigned more synthetic samples to be generated. To avoid generating noise, Kmeans-Smote [7] initially segments the minority samples via the K-means clustering method, followed by the application of Smote within each cluster, thereby attaining a balanced dataset. Nromm [25] is proposed to avoid generating noise by eliminating noise within clusters of data samples and developing an adaptive embedding algorithm. In addition, GB-Smote [27], which adopts a sample grouping scheme to make the new samples further from the overlapping region and rectify the decision boundary properly, is also a powerful oversampling method.

Sample generation represents the mechanism of new sample generation using selected initial samples. To mitigate the issue of overdense sample generation, CHSmote [14] strategically places new samples within convex hull areas to achieve a more even distribution of synthetic samples. Additionally, RCSmote [20] confronts the risk of overgeneralization, which stems from oversampling both uninformative and noisy samples, by implementing a sample categorization strategy. This approach not only identifies samples apt for oversampling but also incorporates a safety margin in the accuracy of calculations during the sample generation process. To expand the minority class area, Borderline-Smote2 [18] can generate a portion of new samples between a minority sample and its nearest majority sample. Similarly, SVM-Smote [19] adopts both interpolation and extrapolation strategies to create the new sample based on the density of the majority class sample around qualified minority samples. Geometric-Smote [21] enlarges the area of the minority class by creating synthetic samples in the vicinity of the geometric space surrounding each chosen minority sample. Moreover, the mixup method, proposed by Zhang et al. [32], which is originally developed as a general data augmentation technique for deep learning, can also create new samples by performing a linear interpolation between two random samples from the entire dataset (both the input features and label).

Although the abovementioned oversampling schemes have been demonstrated to overcome the class imbalance problem efficiently, they still have some weaknesses, including (1)

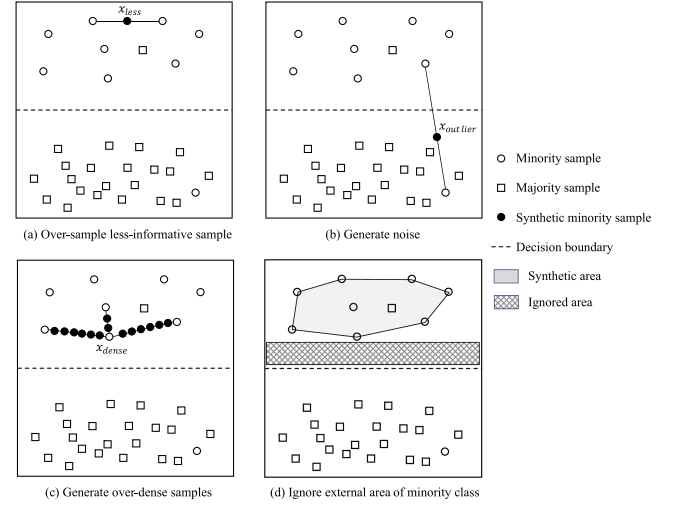


Fig. 1. Four weaknesses resulting in oversampling less-informative minority samples in existing oversampling methods.

generating minority samples far from the boundary, (2) generating noise, (3) generating overly dense synthetic samples, and (4) ignoring the external area of the minority class. The details of these weaknesses are explained in the following.

The first weakness is the generation of less-informative minority samples [14]. Less informative samples represent the samples that are far away from the decision boundary and provide less useful information for classifier. As illustrated in Fig. 1(a), using Smote as an example, a synthetic sample is produced on the line connecting two randomly chosen samples. When these randomly selected samples are distant from the decision boundary, the newly generated samples may also be far from this boundary. Although these distant samples contribute to balancing the datasets, their effectiveness in enhancing the predictive accuracy for the minority class is marginal [14].

The second weakness is generating noisy minority samples [25], which can deteriorate the performance of standard classifiers [22]. In the original dataset, there may be single minority samples in the majority area, which can be noise or outliers. While these examples can often be outliers representing a rare but valid sub-concept, noisy samples can be generated by Smote between an outlier and another original sample from a different sub-concept. As shown in Fig. 1(b), a new noisy minority sample can be generated by the outlier in the majority class area. Moreover, Smote may also generate noise due to multiple subclusters [14]. When the two minority samples come from different subclusters, a new sample is generated between these two samples, and the sample may be noisy since the area between two minority subclusters belongs to the majority class.

The third weakness is generating overly dense synthetic samples [24]. Some oversampling methods (e.g., Smote, Borderline Smote and SVM-Smote) only generate new samples along the line segment between two minority samples. The limited range for generating samples could lead to an excessive concentration of synthetic samples along a line segment, particularly in cases of severe imbalance or when the count of nearest neighbors is

TABLE I
SUMMARY OF WEAKNESSES OF EXISTING OVERSAMPLING METHODS

Smote variants	Weakness 1: Less-informative samples	Weakness 2: Noisy samples	Weakness 3: Dense samples	Weakness 4: External area ignorance
Smote [17]	×	×	×	×
Borderline-Smote1 [18]	✓	✓	×	×
Borderline-Smote2 [18]	✓	✓	×	✓
SVM-Smote [19]	✓	✓	×	✓
Adasyn [23]	✓	✓	×	×
Kmeans-Smote [7]	×	✓	×	×
Geometric Smote [21]	×	✓	✓	×
RCSmote [20]	✓	✓	×	×
CHSmote [14]	✓	✓	✓	×
GB-Smote [27]	✓	✓	×	×

Note: “✓” and “×” denote the corresponding over-sampling method can and cannot overcome the corresponding weakness, respectively.

low [14]. As shown in Fig. 1(c), the newly generated minority samples around x_{dense} are overdense. Consequently, the distribution of minority samples becomes non-uniform. Additionally, instances of synthetic samples overlapping occur when two initial minority samples are nearest neighbors to one another. Such overlapping is of limited utility, as redundant samples seldom introduce new information and are likely to lead to overfitting.

The fourth weakness is ignoring the external area of the minority class. Since some oversampling methods generate synthetic samples along the line segment between two minority samples, the new samples are limited in the internal area of the minority class. As shown in Fig. 1(d), the new samples can only be generated in the area fenced by the borderline samples of the minority class but cannot be generated in the external area between the decision boundary and the borderline of the minority class. Since the ignored area in Fig. 1(d) is closer to the ideal decision boundary [22], the new samples generated inside are difficult to learn and can provide more information for classification. Thus, oversampling minority samples to expand the borderline of the minority class toward the decision boundary may overcome the class imbalance problem better.

Table I lists the cases of whether existing oversampling methods consider the above four weaknesses. While some methods can overcome some weaknesses, they cannot consider these weaknesses simultaneously, which can cause the degradation of classification performance. Instead, our proposed approach aims to simultaneously address all four identified weaknesses.

III. METHODS

A. A New Oversampling method: EB-Smote

In this section, we describe the theory of our proposed EB-Smote. The key idea in EB-Smote is to generate more informative synthetic samples around the borderlines of minority and majority classes, and the borderlines can be expanded toward the decision boundary. EB-Smote consists of two components: (1) selection of border samples and (2) generation of expanding samples. The selection of border samples can address the first two weaknesses, and the generation of expanding samples can address the last two weaknesses explained above.

The first component, selection of seed samples, can address two weaknesses of Smote, i.e., generating less-informative minority samples and generating noise. Initially, in the first component, minority or majority class samples are classified into three categories, i.e., border, outlier, and safety. The border samples are close to the borderline, the outlier samples are the isolated minority class samples existing within the majority class area, and the safety samples are situated within homogeneous regions exclusively occupied by instances belonging to a single class. To avoid generating noise, the outlier samples are removed in the subsequent generation process. To avoid generating less-informative minority samples, we only focus on the border samples around the borderline to generate new synthetic samples. Thus, EB-Smote can generate synthetic samples, providing more information for classification and avoiding generating noise.

The second component, the generation of expanding synthetic samples, can address another two weaknesses, i.e., generating overly dense synthetic samples and generating synthetic samples in limited areas. To avoid generating overly dense minority samples, new synthetic samples are generated in convex hulls consisting of multiple samples, which are larger than the line segment between two samples. Thus, the new samples can be more variant. Furthermore, to avoid generation in the limited internal area of the minority class, new synthetic samples are generated between minority and majority samples. Thus, the new samples can be located in the external area that is closer to the ideal decision boundary and thus provide more information for classification.

1) *Selection of Border Samples*: Classifier algorithms typically aim to define the decision boundary between dominant and less represented classes with high precision. Samples that lie on the intersection of class boundaries tend to have a higher risk of being incorrectly classified due to their close spatial relation to the decision boundary [14]. Consequently, such samples at the borderline require extra focus during classification. Conversely, samples distanced from the decision boundary are likely to provide minimal relevant data, thus contributing insignificantly to the classification accuracy. Moreover, the samples invading the area belonging to the other class can be considered noise and provide error information for classification. Therefore, we focus

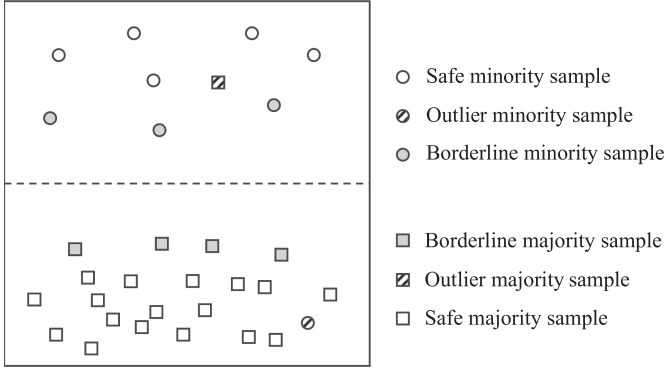


Fig. 2. Selection of border minority and majority samples.

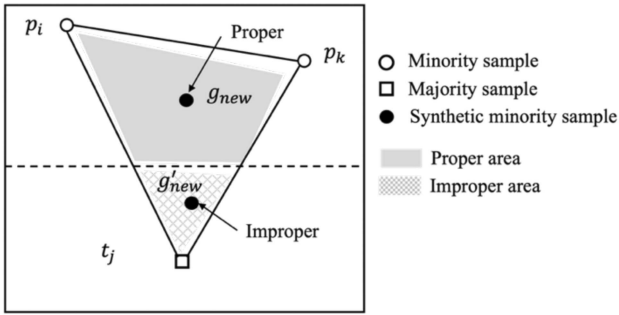


Fig. 3. Proper synthesis area of new minority samples.

on the border samples. As shown in Fig. 2, minority or majority class samples can be classified into three classes, and we only use the border samples to generate new samples.

To separate a sample into border, safe or outlier, we mainly consider its nearest neighbors containing how many samples from the opposite class. Specifically, suppose that the whole training dataset is W , including the minority class $P = \{p_1, p_2, \dots, p_x\}$ and the majority class $T = \{t_1, t_2, \dots, t_y\}$. x and y represent the number of minority and majority examples, respectively. For every $p_i (i = 1, 2, \dots, x)$ in the minority class P , its m_α nearest neighbors from the whole training set W are calculated. Among the m_α nearest neighbors, the number of majority examples is denoted by n_α . If $n_\alpha = m_\alpha$, p_i is classified as outlier. If $m_\alpha/\lambda \leq n_\alpha < m_\alpha$, where λ is a parameter that needs to be predefined, p_i is classified as border. If $0 \leq n < m_\alpha/\lambda$, p_i is safe. The separation process for the majority class is similar. Finally, we can obtain a set of border minority samples $B_{minority} = \{p'_1, p'_2, \dots, p'_{b\alpha}\}$ and a set of border majority samples $B_{majority} = \{t'_1, t'_2, \dots, t'_{b\beta}\}$. The main process of selecting $B_{minority}$ and $B_{majority}$ is detailed in Algorithm 1.

2) *Generation of Expanding Synthetic Samples*: As mentioned earlier, most of the oversampling methods generate synthetic minority samples by interpolating along the line segments connecting pairs of minority class samples. This practice may confine the newly generated minority class samples to the original minority class region. To generate synthetic samples that

Algorithm 1: Selection of the Border Minority and Majority Samples.

Input:

- Imbalanced training dataset W consists of minority class $P = \{p_1, p_2, \dots, p_x\}$ and majority class $T = \{t_1, t_2, \dots, t_y\}$
- Number of nearest neighbors for minority and majority class m_α, m_β
- Threshold for selection of border samples λ

Output:

- Border minority samples $B_{minority}$
- Border majority samples $B_{majority}$

Begin:

for p_i in P do

Find the m_α nearest neighbors from W

Denote the number of majority samples as n_α

If $m_\alpha/\lambda \leq n_\alpha < m_\alpha$:

Put p_i into $B_{minority}$

end

for t_i in T do

Find the m_β nearest neighbors from W

Denote the number of majority samples as n_β

If $m_\beta/\lambda \leq n_\beta < m_\beta$:

Put t_i into $B_{majority}$

end

• **End**

can expand the area of the minority class, an ideal method is to generate new samples between minority samples and majority samples. Since the new synthetic sample can provide more information for the classification when it is closer to the decision boundary, the new sample placed between the border minority and majority samples can be more informative. During this process, two crucial problems need to be addressed, i.e., the first is to avoid generating overdense samples, and the second is to avoid generating noise samples invading the area belonging to the opposite class.

If the new samples are generated along the line segment between a minority sample and a majority sample, the new sample can be overly dense, such as Smote. To tackle this issue, we embrace the approach presented in [14] by constructing convex hulls that encapsulate the local spatial distribution using a selection of samples. Consequently, the synthetic sample generation expands from mere line segments to encompass broader spatial geometric areas. Simply, the new minority samples are generated in the convex hulls based on only two minority samples and a majority sample. Specifically, for each border minority sample p_i in $B_{minority}$, a majority sample t_j is randomly selected from its v_α nearest neighbors from $B_{majority}$, and another border minority sample p_k is randomly selected from t_j 's w_α nearest neighbors from $B_{minority}$. Then, a new minority sample g_{new} is generated in the convex hulls based on p_i, p_k, t_j , as shown in Formula (1). The process for the majority

Algorithm 2: Generation of Synthetic Minority and Majority Samples.**Input:**

- Border minority and majority samples $B_{minority} = \{p'_1, p'_2, \dots, p'_{b_\alpha}\}, B_{majority} = \{t'_1, t'_2, \dots, t'_{b_\beta}\}$
- Ratio between minority samples and majority samples after resampling r
- Expand degree of minority and majority class $Expand_\alpha, Expand_\beta$
- Number of nearest neighbors in oversampling $v_\alpha, w_\alpha, v_\beta, w_\beta$

Output:

- Generated new minority samples $G_{minority}$
- Generated new majority samples $G_{majority}$

Begin:

for each t_i in $B_{majority}$ do

 Randomly select one sample p_j from t_i 's v_β nearest neighbors in $Border_{minority}$

 Randomly select one sample t_k from p_j 's w_β nearest neighbors in $Border_{majority}$

 Select a random number φ between 0 and 1

 Generate a new sample $a_{new} = (1 - Expand_\beta)(\varphi \cdot t_i + (1 - \varphi) \cdot t_k) + Expand_\beta \cdot p_j$

 Put a_{new} in $G_{majority}$

end

Calculate the number of new minority samples need to generate by each border minority samples

$$g_{num} = (r \cdot (\beta + b_\beta) - \alpha) / b_\alpha$$

for each p_i in $B_{minority}$ do

 Randomly select g_{num} samples $T_{minority}$ from p_i 's v_α nearest neighbors in $Border_{majority}$

 For each t_j in $T_{minority}$:

 Randomly select one sample p_k from t_j 's w_α nearest neighbors in $Border_{minority}$

 Select a random number φ between 0 and 1

 Generate a new sample $g_{new} = (1 - Expand_\alpha)(\varphi \cdot p_i + (1 - \varphi) \cdot p_k) + Expand_\alpha \cdot t_j$

 Put g_{new} in $G_{minority}$

end

End

class is similar.

$$\begin{aligned} g_{new} &= \alpha_1 p_i + \alpha_2 p_k + \alpha_3 t_j \\ \alpha_1 + \alpha_2 + \alpha_3 &= 1 \\ \alpha_1, \alpha_2, \alpha_3 &> 0 \end{aligned} \quad (1)$$

Fig. 3 shows the convex hull based on p_i, p_k, t_j , which is the triangular area. It is worth noting that the new synthetic minority sample can be proper when it is generated in the area of the minority class as g_{new} and can also be improper when it invades the area of the majority class as g'_{new} . The improper synthetic minority sample is noise.

Thus, it is important to avoid generating noise samples invading the area belonging to the opposite class. To address this issue, we improve the generation process as Formula (2).

$$g_{new} = (1 - Expand_\alpha)(\varphi \cdot p_i + (1 - \varphi) \cdot p_k) + Expand_\alpha \cdot t_j \quad (2)$$

where φ is a random number between 0 and 1 and $Expand_\alpha$ is the key parameter between 0 and 1 to avoid generating noise. The higher $Expand_\alpha$ is, the closer the new synthetic minority sample is to the area of the majority and expresses an aggressively expanding degree. As $Expand_\alpha$ is equal to 0, the new synthetic minority sample is generated along the line between two minority samples p_i, p_k . As $Expand_{minority}$ is equal to 1, the new synthetic minority sample overlaps with the majority sample t_j . Thus, by adjusting $Expand_\alpha$, we can properly expand the borderline of the minority class. Similarly,

the process of the majority class is similar. Finally, we can obtain the new synthetic minority and majority samples $G_{minority}$ and $G_{majority}$. The principal procedure for generation is elaborately outlined in Algorithm 2.

3) *Analysis of EB-Smote*: Compared with common oversampling methods, the two key components in EB-Smote can alleviate the class imbalance problem better. The first component, selection of border samples, can address two weaknesses of Smote, i.e., generating less-informative minority samples and generating noise. After separating the samples into border, noise and safety, the next generation process only focuses on the border samples around the borderline. Thus, EB-Smote can generate synthetic samples, providing more information for classification and avoiding generating noise.

The second component, the generation of expanding synthetic samples, can address another two weaknesses of Smote, i.e., generating overly dense synthetic samples and generating synthetic samples in a limited area. Since the new synthetic samples are generated in the convex hulls consisting of three samples but not the line segment between two samples, the overdense problem is alleviated. Furthermore, new synthetic samples are generated between minority and majority samples and thus can expand the sample area and drive the original borderline closer to the ideal decision boundary.

In addition to generating minority samples, EB-SMOTE also generates majority samples around the borderline. In other oversampling methods, it is intuitive to oversample the minority

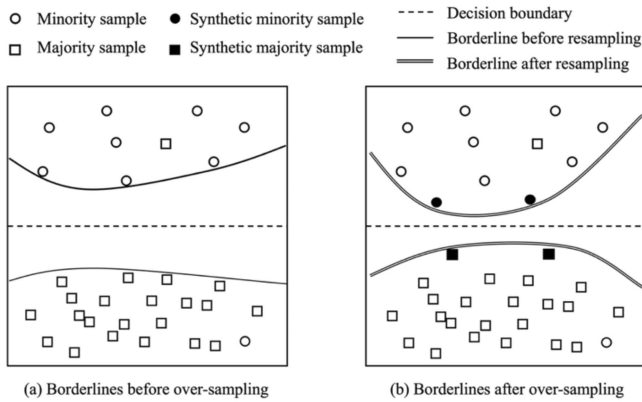


Fig. 4. Comparison of borderlines before and after oversampling by EB-Smote.

class samples instead of majority class samples to balance the class distribution. Since oversampling majority class samples can increase the sample size of the majority class and exacerbate the class imbalance problem, existing methods usually overlook the generation of majority samples. However, the decision boundary of the classification model is constructed based on both minority and majority samples. Oversampling the majority samples around the borderline can also be helpful to optimize the decision boundary.

The ideal resampled dataset distribution based on EB-Smote is shown in Fig. 4. Comparing the borderlines before Fig. 4(a) and after (Fig. 4(b)) oversampling, we find that the new synthetic minority and majority samples expand their respective borderlines toward the ideal decision boundary. Hence, the newly generated synthetic samples, compared to those created within the original area, can offer more valuable information for classification, thereby more effectively mitigating the issue of class imbalance.

It is worth noting that before applying BE-Smote, 10 important parameters need to be predefined, as shown in Table II. The setting of these parameters is determined based on the grid search method, of which the main idea is to select the best parameters after comparing the performance of models with different parameters.

B. Prediction Models

To evaluate the performance of our proposed oversampling method EB-Smote, we need to construct prediction models that can process the resampled training dataset. We use three machine learning models, which are commonly used in previous studies, including decision tree (DT), support vector machine (SVM), and artificial neural network (ANN) [33], [34]. We specifically concentrate on these fundamental classifiers, excluding more complex ones such as ensembles, as they involve multiple elements that could be influenced by data characteristics in imbalanced datasets [22]. Additionally, we omit the consideration of the logistic regression classifier, as the decision boundary between minority and majority classes is not linear in most cases.

TABLE II
PARAMETERS NEED TO BE PREDEFINED IN BE-SMOTE

Parameter	Meaning
m_α, m_β	Number of nearest neighbors to select border minority and majority samples
λ	thresholds control the selection of border samples
r	Ratio between minority samples and majority samples after resampling
$Expand_\alpha$	Expand degree of minority class during generation
$Expand_\beta$	Expand degree of majority class during generation
v_α, v_β	Number of nearest neighbors to select opposite class samples
w_α, w_β	Number of nearest neighbors to select same class samples

SVM is a robust and distribution-free machine learning method [35]. SVM operates as a generalized linear model, establishing an optimal hyperplane to serve as the decision boundary. This hyperplane is devised to maximize the margin between itself and the nearest training samples, simultaneously ensuring accurate classification. The training samples situated closest to this optimal hyperplane are termed support vectors, with the remainder of the training samples having no influence on the determination of the optimal hyperplane. SVM's learning methodology revolves around resolving a convex quadratic programming challenge. For samples that are not linearly separable, SVM employs nonlinear kernel functions to project input vectors into a higher-dimensional feature space, thereby enhancing separability. To curtail overfitting, the penalty for misclassification is fine-tuned.

A decision tree is a type of predictive model in the form of a tree structure, which possesses stability and interpretability [36]. The decision tree consists of nodes, branches, and leaves. The nodes represent the input variables, and the branches represent the possible values of the input variables. The leaves represent the outcome or decision that is made based on the input variables and their values. The decision tree starts with a single node that represents the entire dataset. The algorithm then selects the input variable that is most useful in splitting the dataset into two or more subsets. The goal is to create subsets that are as homogeneous as possible, meaning they have similar outcomes or decisions. Once the tree is constructed, it can be used to make predictions for new instances by following the branches from the root node to the appropriate leaf node. Importantly, to obtain the generalization performance, we control the depth of the trees and adopt a pruning strategy.

Artificial neural networks, also known as deep learning, are one of the most popular artificial intelligence techniques used in credit default prediction [37]. This model draws inspiration from the neural processing mechanisms of the human brain and comprises an extensive network of interconnected neurons. In

binary classification tasks, a neural network typically features three distinct layers: (1) an input layer, hosting a number of neurons equivalent to the count of explanatory variables, (2) one or more hidden layers, each populated with a specific number of neurons, and (3) an output layer, containing a single neuron. Neurons within a particular layer are linked to those in the subsequent layer, with each linkage characterized by a weight that signifies the strength of the connection between two neurons. The value for each neuron is derived from a weighted sum of its inputs, subsequently processed through a nonlinear activation function. Throughout the training phase, weights are fine-tuned via back-propagation to diminish the disparity between the network's actual outputs and the anticipated outcomes, employing a cost function like cross-entropy for minimization. This procedure continues until a predefined stopping criterion is reached, with classification executed by measuring the output against a threshold. To address overfitting, the dropout technique is utilized, randomly deactivating certain connections during training.

C. Performance Evaluation for Imbalanced Prediction

As for the binary classification problem, we can assess the out-of-sample performance using metrics commonly used for classification models. While the overall accuracy is a widely used metric in classification, it cannot reflect reliable prediction performance when encountering class imbalance problems [19]. This is because the overall accuracy is still high when the classifier misclassifies all majority samples but classifies all the minority samples correctly. Thus, we adopt other more reasonable evaluation metrics.

To assess the predictive performance of the minority and majority classes independently, we distinguish between Type I accuracy (also known as Recall) and Type II accuracy (also referred to as Specificity) [38]. Type I accuracy represents the ratio of correctly predicted minority samples by the model, whereas Type II accuracy denotes the ratio of correctly predicted majority samples by the model. We also utilize the precision metric to evaluate the performance. These metrics are encapsulated in Formulas (3), (4) and (5), respectively.

$$\text{Type I accuracy} = TP / (TP + FN) \quad (3)$$

$$\text{Type II accuracy} = TN / (FP + TN) \quad (4)$$

$$\text{Precision} = TP / (TP + FP) \quad (5)$$

where TP and TN denote the number of minority and majority samples that are classified correctly, respectively, while FN and FP denote the number of misclassified minority and majority samples, respectively.

In addition, area under the precision-recall curve (AUPRC) and the area under the receiver operating characteristic curve (AUROC) are commonly used metrics for estimating the overall performance of an imbalanced dataset [28], [30], [39], [40]. The AUPRC and AUROC are independent of the decision criterion selected and prior probabilities, making them more appropriate for the class imbalance problem [17]. The AUROC measures the trade-off between true positives and false positives, while

the AUPRC measures the trade-off between precision and recall. The AUROC or AUPRC is a portion of the area of the unit square, and therefore, its value ranges from 0 to 1.0. The closer the AUROC or AUPRC is to 1, the better the predictive performance.

IV. EXPERIMENTAL RESULTS

A. Imbalanced Dataset

To validate the efficacy of the proposed EB-Smote algorithm in mitigating the class imbalance issue, we execute experiments on 27 imbalanced datasets sourced from the Public UCI, KDD, and LIBSVM Repository. A summary of these datasets is provided in Table III. These datasets exhibit considerable diversity in their characteristics: the number of features ranges from 7 to 617, and the number of samples varies widely from 360 to 145751. Additionally, the imbalance ratio, calculated as the ratio of majority class samples to minority class samples, spans from 8.60 to 129.53. Leveraging these comprehensive and diverse datasets, including high-dimensional data and a broad spectrum of imbalance ratios, allows for a robust evaluation of various oversampling methods in addressing class imbalance. Further details about these datasets, including data sources and definitions of minority classes, are provided in the Supplement.

B. Prediction Performance of the EB-Smote Method

In this section, we compare the default prediction performance of our proposed oversampling method EB-Smote with other commonly used oversampling methods, including random oversampling, Smote, Adasyn, Borderline1-Smote, Borderline2-Smote and SVM-Smote. Moreover, we also utilize two deep learning based over-sampling methods: Tabular Variational Autoencoder (TVAE) and Conditional Tabular Generative Adversarial Network (CTGAN) [41]. These two methods can generate new minority samples by learning the distributional characteristics of the minority samples. Referring to [19], the number of nearest neighbors that are used in these oversampling methods is set to 5. In EB-Smote, the parameters that need to be predefined are listed in Table III. For comparability, the parameters for the numbers of nearest neighbors are also set to 5, i.e., m_α , m_β , v_α , v_β , w_α , and w_β . The other parameters are determined based on the grid search method.

For all oversampling methods, we defined the search space for the ratio between minority samples and majority samples after resampling as {0.6, 0.8, 1, 1.2, 1.4}. Additionally, for EB-Smote, we explored the $Expand_{minority}$ and $Expand_{majority}$ with values ranging from 0.1 to 0.9 in increments of 0.1, and the λ parameter with values {2, 3, 4}. For the hyper-parameter search space of the machine learning models implemented using the Python package Scikit Learn, we define the hyperparameters for DT with max_depth values of {2, 4, 6, None} and min_samples_split values of {2, 4, 8, 16}. For SVM, we consider the regularization parameter C with values {0.1, 1, 5, 10} and kernel functions of {'linear', 'rbf', 'sigmoid'}. For ANN, the hidden_layer_sizes are set as {(5), (10), (5, 5), (10, 10)}, and learning_rate_init was considered with values {0.001, 0.01,

TABLE III
DESCRIPTION OF THE 27 IMBALANCED DATASETS IN EXPERIMENTS

Dataset	# Features	# Samples	# Majority	# Minority	Imbalance ratio
ecoli	7	336	301	35	8.60
optical_digits	64	5620	5066	554	9.14
satimage	36	6435	5809	626	9.28
pen_digits	16	10992	9937	1055	9.42
abalone	10	4177	3786	391	9.68
sick_euthyroid	42	3163	2870	293	9.80
spectrometer	93	531	486	45	10.80
car_eval_34	21	1728	1594	134	11.90
isolet	617	7797	7197	600	12.00
us_crime	100	1994	1844	150	12.29
yeast_ml8	103	2417	2239	178	12.58
scene	294	2407	2230	177	12.60
libras_move	90	360	336	24	14.00
thyroid_sick	52	3772	3541	231	15.33
coil_2000	85	9822	9236	586	15.76
arrhythmia	278	452	427	25	17.08
solar_flare_m0	32	1389	1321	68	19.43
oil	49	937	896	41	21.85
car_eval_4	21	1728	1663	65	25.58
wine_quality	11	4898	4715	183	25.77
letter_img	16	20000	19266	734	26.25
yeast_me2	8	1484	1433	51	28.10
webpage	300	34780	33799	981	34.45
ozone_level	72	2536	2463	73	33.74
mammography	6	11183	10923	260	42.01
protein_homo	74	145751	144455	1296	111.46
abalone_19	10	4177	4145	32	129.53

0.1}. The remaining hyper-parameters were kept as the default parameters in Scikit Learn.

For comparability, the training and testing process of these methods is the same. The process includes (1) randomly splitting the dataset into training and test datasets with a ratio of 80%:20%, as in previous studies [6], based on stratified sampling for the same ratios between minority and majority in training and test datasets; (2) oversampling the training data; (3) performing grid search with 5-fold cross-validation on the oversampled training data to estimate optimal parameters; (4) training the prediction model; and (5) evaluating the prediction performance in test data. Each experimental procedure is repeated 100 times, utilizing 10 different random seeds for the division of training and test sets, as well as 10 random seeds for the oversampling methods. Finally, the average out-of-sample prediction results of different oversampling methods are summarized.

Due to length constraints, we include detailed results in the Supplement, which describes the classification performance based on different classifiers, oversampling methods, datasets and evaluation metrics. For the readers' convenience, we calculate the ranking of different oversampling methods in descending order based on different evaluation metrics and obtain the average ranking on 27 datasets. As shown in Table IV, the first three columns show the ranking results of different oversampling methods based on DT, SVM and ANN. The final column further shows the average results on DT, SVM and ANN. A smaller value of the ranking denotes better performance of the corresponding oversampling methods.

For Type I accuracy (or Recall) in Table IV, which denotes the rate at which minority class samples are accurately

classified, we find that our EB-Smote method outperforms the other oversampling methods on average. This result means that the resampled training dataset of EB-Smote can contribute more to classifying minority credit default accurately. Moreover, relative to no sampling, any oversampling method can improve the Type I accuracy. This is because the percentage of minority class samples in the training dataset after oversampling is larger than that with no sampling. Then, the larger percentage of minority class samples can emphasize the importance of minority credit default in the prediction model and increase the Type I accuracy.

For Type II accuracy in Table IV, which denotes the rate at which the majority of nondefault samples are accurately classified, we find that no sampling method is better than other oversampling methods on average. However, such good Type II accuracy does not mean that no sampling is better than other methods. In practice, minority class samples are more of an interest than majority class samples. We find that the Type I accuracy of no sampling is so poor, which means that no sampling method sacrifices the relatively important Type I accuracy to obtain good Type II accuracy. It is crucial to find a balance between Type I accuracy and Type II accuracy. Compared with other oversampling methods, our EB-Smote model obtains more balanced results across different models.

For Precision in Table IV, which represents the proportion of the instances the model predicted as positive are actually correctly predicted as positive, we find that our proposed EB-Smote method outperforms the other oversampling methods on average. Given EB-Smote's better performance in both Recall and Precision, it demonstrates that EB-Smote achieves a good balance between identifying minority cases and avoiding false

TABLE IV
AVERAGE RANK OF PREDICTION PERFORMANCE FOR DIFFERENT OVER-SAMPLING METHODS

Panel A: Type I accuracy (or Recall) (%)					Panel B: Type II accuracy (%)				
Oversampling Method	DT	SVM	ANN	Average	Oversampling Method	DT	SVM	ANN	Average
No sampling	7.35	9.46	9.54	8.78	No sampling	3.12	1.62	1.73	2.16
Random over-sampling	7.77	4.54	3.77	5.36	Random over-sampling	2.04	6.69	7.46	5.40
Smote	4.81	4.50	4.96	4.76	Smote	6.54	6.12	6.58	6.41
Adasyn	5.42	3.77	4.77	4.65	Adasyn	6.81	7.73	7.31	7.28
Borderline1-Smote	7.19	5.35	5.23	5.92	Borderline1-Smote	5.38	6.65	6.31	6.11
Borderline2-Smote	6.69	4.12	5.08	5.30	Borderline2-Smote	6.27	7.92	7.96	7.38
SVM-Smote	4.54	6.19	5.35	5.36	SVM-Smote	4.96	4.73	5.58	5.09
TAVE	3.31	6.65	5.96	5.31	TAVE	7.19	3.58	3.81	4.86
CTGAN	4.12	7.73	6.62	6.16	CTGAN_OS	7.27	3.23	4.00	4.83
EB-Smote	2.50	1.38	1.58	1.82	EB-Smote	4.85	5.35	4.04	4.75

Panel C: Precision (%)					Panel D: PRAUC (%)				
Oversampling Method	DT	SVM	ANN	Average	Oversampling Method	DT	SVM	ANN	Average
No sampling	4.85	4.81	3.81	4.49	No sampling	6.62	6.77	7.12	6.84
Random over-sampling	2.77	6.00	6.54	5.10	Random over-sampling	4.31	5.88	5.42	5.20
Smote	6.42	5.08	6.35	5.95	Smote	5.69	5.23	6.08	5.67
Adasyn	6.38	6.58	7.12	6.69	Adasyn	6.42	6.04	6.42	6.29
Borderline1-Smote	6.00	6.00	5.96	5.99	Borderline1-Smote	6.42	6.85	5.81	6.36
Borderline2-Smote	6.42	7.42	7.42	7.09	Borderline2-Smote	7.04	6.96	5.81	6.60
SVM-Smote	4.35	4.50	5.85	4.90	SVM-Smote	4.04	5.31	4.96	4.77
TAVE	6.73	4.27	4.42	5.14	TAVE	5.42	5.92	5.19	5.51
CTGAN	7.38	5.42	4.35	5.72	CTGAN	5.50	2.77	6.15	4.81
EB-Smote	3.65	4.19	3.15	3.66	EB-Smote	3.50	3.15	2.00	2.88

Panel E: AUROC (%)				
Oversampling Method	DT	SVM	ANN	Average
No sampling	6.96	7.08	6.58	6.87
Random over-sampling	7.62	4.54	4.81	5.66
Smote	5.00	4.62	5.85	5.16
Adasyn	5.65	5.38	6.15	5.73
Borderline1-Smote	7.19	6.77	6.77	6.91
Borderline2-Smote	6.57	4.25	6.04	5.62
SVM-Smote	4.35	6.19	5.54	5.36
TAVE	3.42	5.85	4.96	4.74
CTGAN	4.58	6.12	4.65	5.12
EB-Smote	2.96	3.77	3.04	3.26

Note: the bold numbers denote the best rank of prediction performance among different over-sampling methods.

TABLE V
WILCOXON SIGNED-RANK BETWEEN EB-SMOTE (W+) AND OTHER OVER-SAMPLE METHODS (W-) ON ALL SELECTED DATASETS

Compared methods	PRAUC			AUROC		
	R+	R-	p-value	R+	R-	p-value
EB-Smote vs. No sampling	2363	958	<0.0001*	2619	702	<0.0001*
EB-Smote vs. Random over-sampling	2543	778	<0.0001*	2692	629	<0.0001*
EB-Smote vs. Smote	2653	668	<0.0001*	2648	673	<0.0001*
EB-Smote vs. Adasyn	2891	430	<0.0001*	2910	411	<0.0001*
EB-Smote vs. Borderline-Smote1	2854	467	<0.0001*	3017	304	<0.0001*
EB-Smote vs. Borderline-Smote2	3046	275	<0.0001*	3063	258	<0.0001*
EB-Smote vs. SVM-Smote	2532	789	<0.0001*	2618	703	<0.0001*
EB-Smote vs. TVAE	2746	575	<0.0001*	2341	980	0.0014*
EB-Smote vs. CTGAN	3016	305	<0.0001*	2315	1006	0.0021*

minority samples, making it highly effective for addressing the class imbalance problem. In contrast, the performance of the DT model with Random oversample method shows a better Precision ranking but a worse Recall performance. This suggests that while Random oversample is effective at avoiding false positive minority samples, it struggles to accurately identify the true minority samples. For example, if the model classifies only one sample as minority and it is correct, the precision would be 100%. However, this would indicate that many minority samples have still not been identified, leading to a low Recall.

For the PRUAC and AUROC in Table IV, which can comprehensively evaluate the prediction performance, we find that our proposed EB-Smote method outperforms the other methods not only on average but also across most classifiers. This result demonstrates that the resampling strategy in EB-Smote can indeed be more helpful in solving the class imbalance problem. Moreover, compared with the no sampling method, almost all oversampling methods obtain higher AUROC values. This indicates that increasing the number of minority class samples in the training dataset can be helpful for the imbalance problem, regardless of the oversampling strategy.

The reason for the improvement of AUROC is that, with an increased number of minority class examples, the classifier has a greater likelihood of correctly predicting positive instances, resulting in an increase in TP and a reduction in FN. Consequently, TPR improves. Since the synthetic samples generated are added only to the minority class, FPR remains stable or is only slightly affected. Thus, the improvement of TPR without significantly raising FPR will shift the ROC curve upward, thereby increasing the AUROC score. Additionally, EB-Smote generates informative synthetic samples for the majority class, enhancing the classifier's ability to correctly predict negative instances. This results in an increase in TN and a decrease in FP, thereby lowering the FPR. In summary, the increase in TPR shifts the ROC curve upward, while the decrease in FPR shifts it to the left, collectively leading to a higher AUROC score.

Moreover, we conduct tests to ascertain statistically significant differences between EB-Smote and other oversampling techniques. We opt for the nonparametric Wilcoxon signed-rank test due to its suitability over parametric testing methods in scenarios where assumptions of independence, normality, and homogeneity of variances are not met. The outcomes of the Wilcoxon signed-rank test, comparing EB-Smote (R+) against

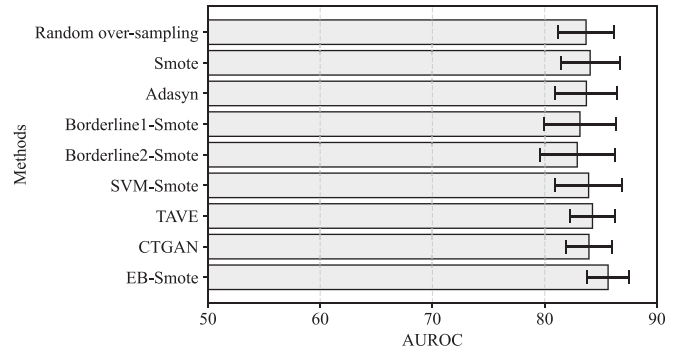


Fig. 5. The error bar of the AUROC performance for oversampling methods.

other oversampling methods (R-) at PRAUC and AUROC metrics, are displayed in Table V. An R+ value exceeding R- signifies superior performance of EB-Smote over the method it is compared with, and the inverse is also true. A star adjacent to the p-value numbers denotes the statistical significance of the difference between the methods compared. Notably, low p-values (<0.05) demonstrate that EB-Smote significantly outperforms other oversampling methods.

Furthermore, since our proposed EB-Smote method involves a stochastic process during the generation new synthetic samples, its stability needs to be investigated. The randomness during the new sample generation is caused by setting different random seeds. Therefore, we calculated the standard deviation of the AUROC performance across 10 experiments with 10 different random seeds. We computed the average standard deviation across different training and testing data splits, imbalanced datasets, and classifiers. The results are shown in Fig. 5, where the length of the bars represents the AUROC, and the error bars indicate the standard deviation. It can be observed that the proposed EB-SMOTE method not only outperforms other methods but also exhibits a smaller AUROC variance. It suggests that the performance of EB-Smote is stable in handling class imbalance problem.

C. Ablation Analysis of the EB-Smote Method

To systematically analyze the effectiveness of components in EB-Smote, we conduct ablation analysis from two aspects, i.e.,

TABLE VI
ABLATION RESULTS OF THE OVERSAMPLING PROCESS FOR EB-SMOTE

Metrics	Over-sample methods	DT	SVM	ANN	Average
Type I accuracy (%)	EB-Smote _{-select}	60.32	61.74	76.16	66.07
	EB-Smote _{-generate}	52.09	54.77	72.07	59.64
	EB-Smote	63.86	70.17	81.91	71.98
Type II accuracy (%)	EB-Smote _{-select}	94.61	91.89	88.25	91.58
	EB-Smote _{-generate}	97.00	96.54	89.28	94.27
	EB-Smote	95.84	97.16	92.63	95.21
Precision (%)	EB-Smote _{-select}	45.28	44.83	41.89	44.00
	EB-Smote _{-generate}	53.31	47.94	42.24	47.83
	EB-Smote	51.11	52.39	49.36	50.95
PRAUC	EB-Smote _{-select}	37.10	52.79	54.69	48.19
	EB-Smote _{-generate}	39.20	52.71	52.90	48.27
	EB-Smote	40.35	55.77	58.79	51.64
AUROC (%)	EB-Smote _{-select}	77.54	88.17	87.62	84.44
	EB-Smote _{-generate}	74.35	88.46	86.45	83.09
	EB-Smote	78.66	89.17	89.11	85.65

Note: the bold numbers denote the best prediction performance based on each metric.

the oversampling process and the oversampling objects. The details are described as follows.

1) *Ablation Results of the Oversampling Process*: During the oversampling process, EB-Smote has two crucial components. identifying the noise, safety and border samples and generating new expanding samples based on samples from both minority and majority classes. To verify the impact of these components in EB-Smote, we compare the original EB-Smote with its two variants, EB-Smote_{-select} and EB-Smote_{-generate}. EB-Smote_{-select} does not initially distinguish samples into outlier, safety and border and then randomly uses samples to generate new samples. EB-Smote_{-generate} generates new samples from the line segments connecting the border samples from only the minority or the majority class. For comparison, the experimental processes of the two variants are consistent with EB-Smote. Table VI summarizes the average performance of the two variants and EB-Smote on the 27 datasets described in Table III.

Focusing on the Type I accuracy in Table VI, we find that both EB-Smote_{-select} and EB-Smote_{-generate} perform worse than EB-Smote. This means that the two components in EB-Smote are important for the identification of minority samples. Furthermore, the decline of EB-Smote_{-generate} is higher than that of EB-Smote_{-select}, which suggests that generating new samples in the convex hull area between minority and majority samples is more important than selecting samples around the borderlines. For Type II accuracy, EB-Smote_{-generate} outperforms EB-Smote_{-select}. The reason is that for EB-Smote_{-generate}, the slight improvement in Type II accuracy actually sacrifices significant Type I accuracy. For Precision, EB-Smote_{-select} and EB-Smote_{-generate} perform worse than EB-Smote, which suggests that these two components are important in accurately predicting minority class samples.

TABLE VII
ABLATION RESULTS OF OVERSAMPLING OBJECTS FOR EB-SMOTE

Metrics	Over-sample methods	DT	SVM	ANN	Average
Type I accuracy (%)	EB-Smote _{-majority}	59.53	61.02	74.52	65.02
	EB-Smote _{-minority}	52.27	22.02	35.50	36.60
	EB-Smote	63.86	70.17	81.91	71.98
Type II accuracy (%)	EB-Smote _{-majority}	94.13	92.22	88.21	91.52
	EB-Smote _{-minority}	97.13	98.86	99.09	98.36
	EB-Smote	95.84	97.16	92.63	95.21
Precision (%)	EB-Smote _{-majority}	44.00	39.93	40.13	41.35
	EB-Smote _{-minority}	53.02	47.87	50.61	50.50
	EB-Smote	51.11	52.39	49.36	50.95
PRAUC (%)	EB-Smote _{-majority}	39.23	52.28	51.39	47.63
	EB-Smote _{-minority}	35.23	50.70	51.24	45.72
	EB-Smote	40.35	55.77	58.79	51.64
AUROC (%)	EB-Smote _{-majority}	76.90	87.77	86.94	83.87
	EB-Smote _{-minority}	74.73	86.15	84.73	81.87
	EB-Smote	78.66	89.17	89.11	85.65

Note: the bold numbers denote the best prediction performance based on each metric.

Focusing on the comprehensive metric PRAUC and AUROC in Table VI, we find that the performance of EB-Smote_{-select} and EB-Smote_{-generate} is lower than EB-Smote. This means that the two components in EB-Smote are crucial for alleviating the class imbalance problem. Compared with EB-Smote_{-select}, the AUROC of EB-Smote_{-generate} is lower. This suggests that the component sample generation in EB-Smote plays a more important role than the component sample selection.

2) *Ablation Results of Oversampling Objects*: For the oversampling objects, EB-Smote oversamples not only the minority class samples but also the majority class samples. To analyze the effects of the newly generated minority and majority samples on the imbalanced problem, we compare the original EB-Smote with two variants that do not sample the minority and majority class samples, denoted as EB-Smote_{-majority} and EB-Smote_{-minority}, respectively. For comparison, the experimental processes of the two variants are consistent with EB-Smote. Table VII summarizes the average performance of the two variants and EB-Smote, on the 27 datasets described in Table III.

In Table VII, at almost all prediction models and on average, we can find that EB-Smote_{-minority} and EB-Smote_{-majority} obtain lower Type I accuracy, Precision, PRAUC and AUROC than EB-Smote. While EB-Smote_{-minority} can obtain higher Type II accuracy than EB-Smote, the sacrifice of Type I accuracy far exceeds the gain of Type II accuracy. These results demonstrate the significance of sampling both minority and majority class samples in the class imbalance problem.

Moreover, the decreased AUROC of EB-Smote_{-majority} is higher than that of EB-Smote_{-minority}, which means that the sampling of minority samples is more important than the sampling of majority samples. In summary, both minority sampling

TABLE VIII
AVERAGE AUROC PERFORMANCE ON IMBALANCED DATASET WITH DIFFERENT OVERLAP DEGREE

Overlap ratio	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%	Average rank
No sampling	93.63	89.70	89.69	83.62	79.15	68.89	61.21	56.28	49.06	97.97	7.6
Random over-sampling	93.59	90.34	90.22	83.48	79.62	70.05	65.16	59.12	49.35	98.03	5.2
Smote	93.98	90.74	90.19	83.61	79.68	69.79	64.03	60.35	48.60	97.62	5.5
Adasyn	94.05	90.78	90.27	83.50	79.17	69.18	64.23	60.38	48.52	97.47	5.4
Borderline1-Smote	93.65	90.88	89.86	83.06	79.33	69.27	64.21	60.74	49.70	97.72	5.6
Borderline2-Smote	93.73	90.77	89.75	82.77	79.34	69.96	64.18	59.95	49.92	98.04	5.5
SVM-Smote	93.84	90.92	90.16	83.35	79.84	69.69	64.22	60.26	49.52	97.78	4.7
TAVE	93.32	90.97	89.13	81.70	74.82	73.27	62.05	53.43	48.62	98.55	7.1
CTGAN	94.04	91.56	89.56	82.51	75.79	73.98	58.64	58.18	51.44	98.32	5.4
EB-Smote	95.51	92.28	85.43	88.01	79.72	73.07	66.33	58.54	54.30	98.07	3.0

and majority sampling are significantly important in EB-Smote since sampling one class of them alone cannot achieve the best results. Specifically, sampling minority classes should be a primary consideration, and further sampling majority classes can better solve the class imbalance problem.

D. Sensitivity Analysis of the Expansion Degree in EB-Smote

The key idea of EB-Smote is to expand the borderline toward the optimal decision boundary by generating new samples around the borderline. To expand the borderline properly, it is crucial to control the expanding degree to avoid overexpanding and generating noise samples. In EB-Smote, $Expand_\alpha$ and $Expand_\beta$ are two crucial parameters that control the degree of expanding borderlines of the minority and majority classes, respectively. The setting of these two parameters is important and can influence the performance of EB-Smote. Thus, in this section, we further analyze how the performance changes at different parameter settings.

1) *Expand Degree of Minority Class*: The expansion degree of minority class $Expand_\alpha$ controls the relative distance between the synthetic minority sample and the original borderline of the minority class. Generally, a higher $Expand_\alpha$ means that the synthetic minority sample is further away from the original borderline of the minority class and closer to the ideal boundary separating the minority and majority classes. However, too high $Expand_\alpha$ may also create improper synthetic minority samples that invade the area of the majority class and are considered noise. Thus, it is crucial to analyze what level $Expand_\alpha$ should be set. The results of the prediction performance with different levels of $Expand_\alpha$ are presented in Fig. 6. There are 12 subplots in Fig. 6, where each column represents an evaluation metric, the first three rows represent three predictive models, and the final row represents the average performance of the three predictive models. Each point on the curve represents the average performance of the corresponding classifier on the 27 datasets described in Table III.

Focusing on the first column of the Type I accuracy in Fig. 6, as $Expand_\alpha$ increases, the curve of the average Type I accuracy generally tends to increase first and then decline slightly.

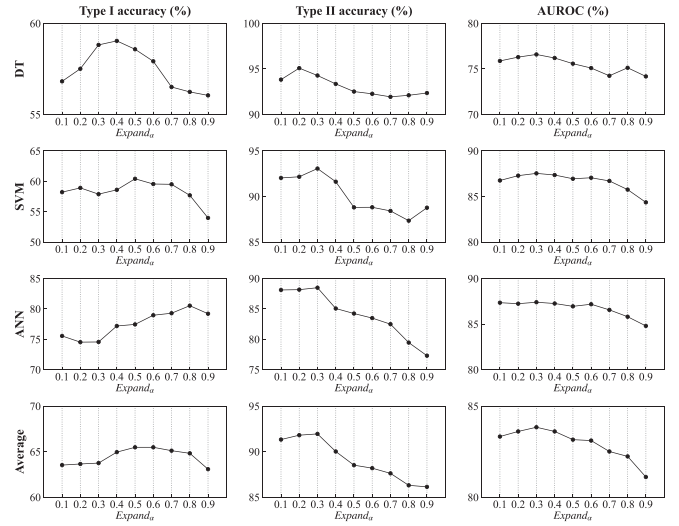


Fig. 6. Performance with different levels of $Expand_\alpha$.

This means that properly generating minority samples that are closer to the decision boundary can provide more information for accurately classifying the minority samples. The Type I accuracy achieves the best performance when $Expand_\alpha$ equals 0.5. However, this does not mean that setting $Expand_\alpha$ to 0.5 is the best in all scenarios. The reason is that the characteristics of different models and the distribution of different datasets vary. As shown in Fig. 6, the Type I accuracy curves of different models present different trends.

Focusing on the second column Type II accuracy in Fig. 6, as $Expand_\alpha$ increases, we find that the curves tend to increase slowly first and decrease rapidly across all models and the average performance reaches its maximum value at 0.3. The degradation of performance is because EB-Smote with high $Expand_\alpha$ can generate improper synthetic minority samples that invade the area of the majority class, and the synthetic noisy majority samples can confuse the identification of the original majority samples. Associated with the increasing Type I accuracy as $Expand_\alpha$ increases, the decreasing Type II accuracy means that the new synthetic samples improve the identification

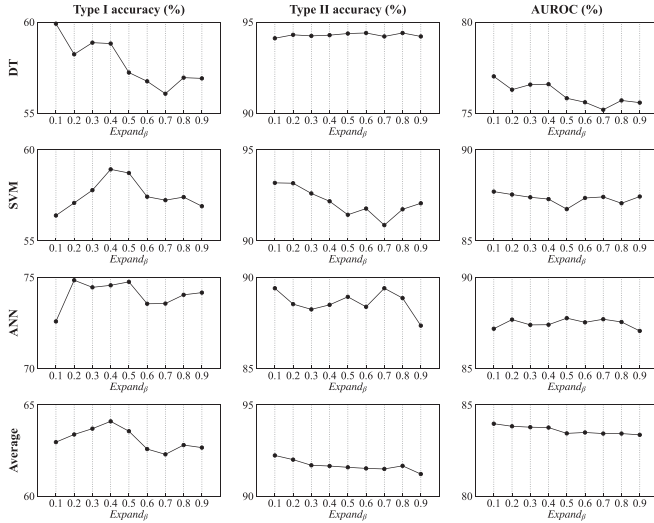


Fig. 7. Performance with different levels of $Expand_\beta$.

of minority samples along with sacrificing the classification of majority samples.

Focusing on the third column AUROC in Fig. 6, as $Expand_\alpha$ increases, the curve of the average AUROC increases slightly until $Expand_\alpha$ is equal to 0.3 and then decreases. However, the curves of different models present different trends. Thus, in practice, we should set different $Expand_\alpha$ for different models properly to obtain better performance.

2) *Expand Degree of Majority Class*: Similarly, the expansion degree of the majority class $Expand_\beta$ controls the relative distance between the synthetic majority sample and the original borderline of the majority class. The results of the prediction performance with different levels of $Expand_\beta$ are presented in Fig. 7.

In Fig. 7, as $Expand_\beta$ increases, across different models, the curves of Type I accuracy, Type II accuracy, and AUROC present different trends. The average Type I accuracy tends to increase at the beginning and then decrease when $Expand_\beta$ is equal to 0.4. Instead, the average Type II accuracy tends to decrease slightly across different $Expand_\beta$. Focusing on the AUROC metric, average performance also decreases slightly. Compared with the characteristics of $Expand_\alpha$ in Section IV.D.1, the overall impact of $Expand_\beta$ on the EB-Smote is relatively small. This is primarily because, to achieve sample balance in the training set, the number of new majority class samples generated during the sampling process is significantly fewer than the number of minority class samples.

E. The Performance of EB-Smote in Imbalanced Datasets With Overlap Problem

In classification tasks with imbalanced datasets, the overlap problem, where instances of the minority class overlap significantly with those of the majority class, has a negative effect on the classifier's performance [13]. This overlap creates an ambiguous boundary between the majority and minority classes, causing the model to confuse minority samples with majority

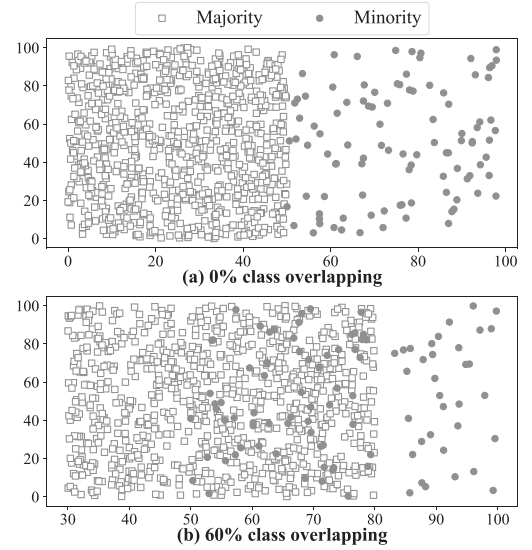


Fig. 8. The example of two different levels of class overlapping.

ones within these overlapping regions. Although commonly used oversampling methods like Smote address class imbalance by generating synthetic minority samples, they often overlook the unique challenges introduced by overlapping areas. Therefore, in this section, we further investigate whether our proposed EB-Smote oversampling method can outperform existing methods on imbalanced datasets with overlap problem.

To assess the performance of oversampling methods across a full spectrum of class overlap levels, we employ artificially generated datasets with varying overlap degrees, ranging from 10% to 100%. Each dataset consists of 1000 samples, with 900 majority samples and 100 minority samples, maintaining an imbalance ratio of 9:1. Each sample contains two features, x and y , where y is randomly distributed within the range $[0, 100]$. For x , minority samples are randomly distributed in the range $[50, 100]$ across all overlap levels, while majority samples are generated in $[0, 50]$ for 0% overlap, $[10, 60]$ for 10% overlap, ..., and $[50, 100]$ for 100% overlap. Fig. 8 illustrates two example datasets with 0% and 60% overlap. It can be observed that the minority class is under-represented in the overlap region.

Consistent with the experimental processes in Section IV.B, we evaluate various oversampling methods across datasets with different overlap degrees based on DT, SVM, and ANN classifiers. Measuring the average AUROC score of three classifiers, Table VIII summarizes the performance on test datasets. We can find that when the degree of overlap degree is low, and all oversampling methods achieve satisfactory AUROC performance, since the decision boundary between majority and minority classes is relatively clear. As the overlap increases, the effectiveness of various oversampling methods declines. When the overlap between majority and minority samples reaches 100%, the AUROC value drops to around 50%, indicating that overlap issues in imbalanced datasets significantly affect classifier performance.

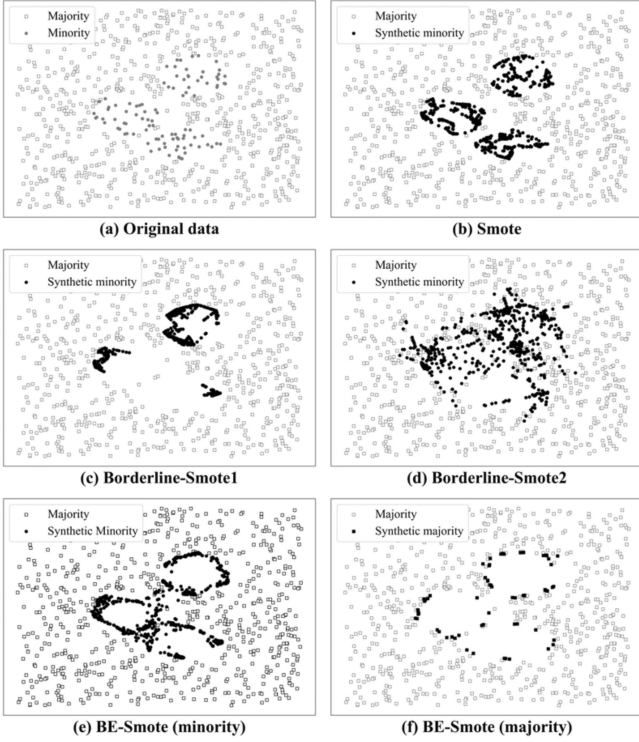


Fig. 9. Sample distribution of the paw02a-800-7-0-BI dataset before and after oversampling.

To further compare the performance of different oversampling methods, we calculated the ranking of each oversampling method across different datasets, with the average ranking presented in the last column of the Table VIII. We can find that our proposed EB-Smote method has the highest average ranking 3.0, outperforming other methods, suggesting that EB-Smote better alleviates the overlap issue in imbalanced datasets compared to other methods. The reason might be that EB-Smote focuses on over-sampling specifically within the border regions of the feature space. By over-sampling border samples, EB-Smote increases the density of minority samples exactly where overlap is most problematic. Additional minority samples around the boundary can help reinforce the minority presence in these overlapping regions. This reinforcement helps the classifier build a decision boundary that is less biased towards the majority class, and more robust in ambiguous areas.

F. Visualization of Different Oversampling Results

To better understand the oversampling results of EB-Smote, we visually compare the distribution of samples in the datasets oversampled by different methods. Due to length constraints, EB-Smote is compared with the popular oversampling methods Smote, Borderline-Smote1, and Borderline-Smote2 [14]. Fig. 9(a) shows the original imbalanced paw02a-800-7-0-BI dataset, collected from KEEL Repository. For clearer visualization, the original minority samples are not visualized in Fig. 9(b)–(f). Compared with the synthetic minority samples by Smote in Fig. 9(b), the synthetic minority samples by the



Fig. 10. Sample distribution of the 04clover5z-800-7-0-BI dataset before and after oversampling.

proposed EB-Smote in Fig. 6(e) are closer to the borderline, which can push the decision boundary closer to the actual decision boundary and provide more information for optimizing the decision boundary.

In addition, to verify that the synthetic samples around the borderline generated by EB-Smote are more reasonable, we visualize other oversampling methods that pay attention to the borderline samples similarly, i.e., Borderline-Smote1 and Borderline-Smote2. As depicted in Fig. 9(c), Borderline-Smote1 emphasizes only a restricted border region, overlooking several critical border areas. In contrast, EB-Smote can cover a broader border area to better help the classifier learn the decision boundary. As shown in Fig. 9(d), Borderline-Smote2, which can generate new minority samples between a minority sample and its nearest majority sample, generates a number of noisy minority samples that invade the area of the majority class. In contrast, EB-Smote avoids generating noise by controlling the expanding degree of synthetic minority samples.

Moreover, a noticeable feature of EB-Smote is that it can also generate majority samples. As shown in Fig. 8(f), the synthetic majority samples are distributed around the borderline of the majority class. As demonstrated in Section IV.C.2, generating majority samples is one of the crucial components in EB-Smote. In conclusion, EB-Smote achieves a more rational sample distribution within the balanced dataset, and thus, EB-Smote can better address the issue of class imbalance.

Furthermore, Figs. 10 and 11 illustrate the sample distributions within the datasets 04clover5z-800-7-0-BI and 03subc15-800-7-0-BI, collected from KEEL Repository. Comparative analysis reveals outcomes similar to those depicted in Fig. 9,

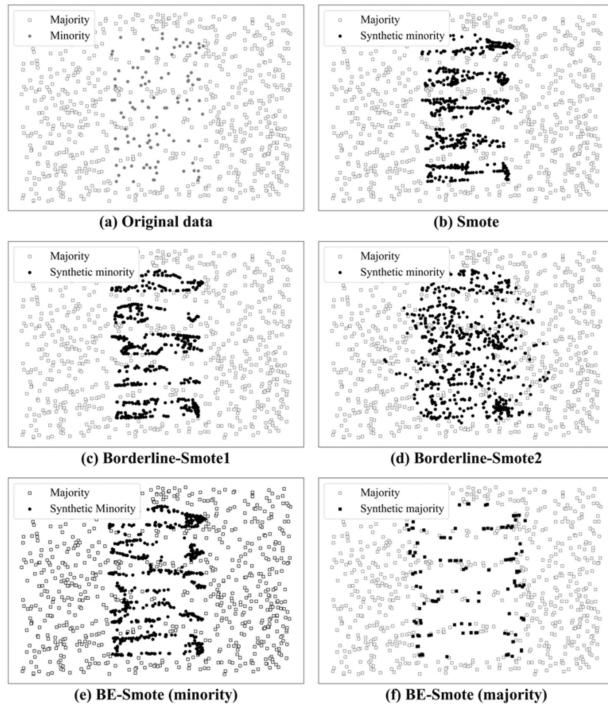


Fig. 11. Sample distribution of the 03 subcl5-800-7-0-BI dataset before and after oversampling.

demonstrating that EB-Smote leads to a more logical distribution of samples in the balanced dataset.

V. CONCLUSION

While machine learning models are increasingly popular in numerous real-world applications, these models encounter the nontrivial challenge of class imbalance. To address the class imbalance problem, this study proposes a novel oversampling method EB-Smote to balance the class distribution by generating synthetic informative samples. EB-Smote consists of two crucial components, i.e., sample selection and sample generation, which can be helpful to generate informative minority and majority class samples and thus better overcome the imbalance problem.

Based on 27 imbalanced datasets, the experimental results demonstrate that EB-Smote can achieve significantly better performance than 8 existing popular oversampling methods. The ablation analysis has verified that two components (i.e., sample selection and sample generation) in EB-Smote both contribute to improving classification performance and that oversampling both minority samples and majority samples can obtain better performance.

This study can provide both theoretical guidance and practical recommendations for solving the crucial class imbalance problem in machine learning. Nonetheless, this paper is not without limitations. For example, our proposed oversampling method has been demonstrated to be helpful in solving the class imbalance in binary classification. However, in other application scenarios, the class imbalance problem exists in multiclass classification. Future studies can further modify our method to

overcome the class imbalance problem in multiclass classification.

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